

■ A.2.5 Recognizing Algebraic Numbers

If you are told that a particular number is the square root of an integer, then given a sufficiently accurate numerical approximation, you can easily recognize what integer the number is the square root of. It is possible to generalize this kind of “number recognizing” to work for numbers that appear as the roots of arbitrary polynomial equation. Here is a package that finds the minimal polynomial equation for which a particular number is a good approximation to a solution.

The heart of the package is a single call to the lattice reduction function `LatticeReduce`. Wrapped around this call are a variety of other functions, which prepare the input, and process the output. Notice the use of indentation to make a sequence of deeply-nested functions slightly easier to read.

```
Recognize::usage = "Recognize[x,n,t] finds a polynomial of degree at
                    most n in the variable t which is satisfied by the number x."
```

```
Recognize[x_Real,n_Integer?Positive,t_] :=
  Block[ {i},
    Table[t^i, {i, 0, n}] .
      Drop[ LatticeReduce[
        Transpose[
          Append[
            IdentityMatrix[n+1],
            Round[
              Table[x^i, {i, 0, n}]
              10^Precision[x]
            ]
          ]
        ]
      ]
    ] [[1]],
    -1
  ]
```

Recognizing algebraic numbers.

Read in the number recognizer.

```
In[1]:= <<Recognize.m
```

This gives the numerical value of $\sqrt{2} + \sqrt{5}$ to 30 decimal places.

```
In[2]:= N[Sqrt[2] + Sqrt[5], 30]
Out[2]= 3.65028153987288474521086239294
```

Recognize finds that this number is a root of a simple fourth-degree polynomial.

```
In[3]:= Recognize[%, 4, x]
Out[3]= 9 - 14 x^2 + x^4
```

NRoots verifies the result.

```
In[4]:= NRoots[%==0, x]
Out[4]= x == -3.65028 || x == -0.821854 ||
        x == 0.821854 || x == 3.65028
```